EEL - USP

LOB 1036 - Geometria Analítica

Lista de exercícios 1 - Parte 1
2º semestre de 2025

1.

a) $P(4,3,-2)$

b) $P(1,9,2)$

c) $k=-2$

d) $P(0,-2,-1)$

2. $\pi: 2x-3y-z-9=0$

3. $\pi: 2x+y-z-1=0$

4. $\pi: 4x+4y+2z+3=0$

5. $\pi: 3x+2y-6=0.$

6. $z=3$

7.

a) $\pi: 4x+5y+3z-6=0$

b) $\pi: x=0$

8. $\alpha=-30$

9. $\pi: y+2z+4=0$

10. $\pi: x-12y-10z-5=0.$

11. $\pi: 2x-8y+3z=0$

12.

a) $\pi: 5x-4y-3z-6=0$

b) $\pi: 2x+2y+z+2=0$

13.

a) $\pi : x+y-2=0$

b) $\pi : 6x-2y+z+3=0$

c) $\pi : y+2z=0$

14. $\begin{cases} x=t \\ y=h \\ z=3t-2h-6 \end{cases}$

15.

a) $\theta = 60^{\circ}$

b) $\theta = 30^{\circ}$

c) $\theta = \cos^{-1} \dfrac{2}{\sqrt{13}}$

d) $\theta = \cos^{-1} \dfrac{3}{\sqrt{13}}$

16. $a = \dfrac{16}{3}, b = \dfrac{20}{3}$

17. $m = \dfrac{1}{2}$

18. $\theta = 60^{\circ}$

19. $\begin{cases} y=-3x+5 \\ z=2x \end{cases}$

20. $r : \begin{cases} x=-1+2t \\ y=-3t \\ z=7t \end{cases}$

21. $m=-2, n=3$

22.

a) $\begin{cases} y=x-2 \\ z=1-2x \end{cases}$

b) $r : \begin{cases} y=2+2x \\ z=3 \end{cases}$

23. $I(2,1,0), I\left(\dfrac{3}{2},0,\dfrac{1}{2}\right), I(0,-3,2)$

24. $I(2,0,0), I(0,1,0), I(0,0,-4), \quad r : \begin{cases} y=-\dfrac{x}{2}+1 \\ z=0 \end{cases}$

25. $\pi : 5x - 7y - 10z = 0$

26. $r : \left\{ \frac{x}{1} = \frac{y}{2} = \frac{z-1}{1} \right.$